

DANIEL NGUYEN

Computer Engineering (Co-op)

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WORK EXPERIENCE

Intern Solutions Engineer (Co-op)

Teldio Corporation

 May 2026 – Aug 2026  Ottawa, ON

- Built **Project Echo**, a capture-and-replay tester that records live protocol traffic from integration modules and replays it as automated test input, removing the physical-hardware bottleneck from QA; extended to MOTOTRBO and Modbus and wired into the test suite with TestRail/Jira reporting. Still in use after the term
- Stood up four C-CURE 9000 VMs spanning vendor versions and integrated over both the REST API and the CrossFire .NET SDK, bridged into Go modules with byte-identical events; fixed defects that only appeared live (an IIS session leak, a cross-timezone journal failure) and handed the PoC to R&D
- Shipped and demoed the Avigilon Unity Access module with an MQTT/JSON/TLS event relay (10 actions, 14 event types, 4 transports, 121 tests); found an MQTT 3.1.1 conformance defect in the partner's relay client, reported upstream to Motorola
- Integrated proprietary intrusion panels, DSC PowerSeries Neo (ITv2) and Bosch B4512 (Mode 2), from protocol research to proof of concept; Bosch ahead of schedule with a 93-item test checklist and a PoC linking alarm events to a camera system
- Wrote a Python tool that decrypts and decodes live panel traffic when standard tooling was unavailable; stood up LenelS2 OnGuard and Pro-Watch labs on Proxmox/VMware; authored 8+ artifacts (feasibility reports, SRS, dev logs, QA handoff) and led a QA/SE enablement session

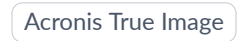
      
   

Computer Technician (Co-op)

XPC

 Mar 2023 – Jun 2023  Toronto, ON

- Diagnosed and repaired 50+ desktops, laptops, and servers with a ~90% recovery rate; built and optimized 20+ custom PCs for gaming and audio-production clients

EDUCATION

BASc in Computer Engineering (Co-op)

University of Ottawa

 2024 – 2029 (expected)  Ottawa, ON

- Third year, CGPA 6.95
- Relevant coursework: Circuit Theory, Digital Systems I & II (VHDL, FPGA), Signal and System Analysis, Computer Architecture I & II, Data Structures & Algorithms, Software Engineering

Currently learning: VHDL/Verilog and FPGA design (Xilinx), assembly, Rust, Kubernetes, LLM agents and MCP

LANGUAGES & INTERESTS

English, Vietnamese · Robotics, embedded systems, computer hardware, gym, volleyball, nutrition

PROJECTS

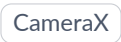
MacTrack

Offline-First Android Nutrition Tracker

 Jun 2026 – Sep 2026

 github.com/Diraclol/MacTrack

An offline-first calorie and macro tracker for Android, built solo in Kotlin and Jetpack Compose. It ships the Canadian Nutrient File on-device, pulls branded foods from Open Food Facts by name or barcode, and scans barcodes offline. Its Room database has survived eight schema migrations without losing a log entry. An optional AI assistant runs on the user's own key, encrypted in the Android Keystore.


Homelab

Self-Hosted Infrastructure

 Jul 2026 – Present

 github.com/Diraclol/Homelab

A self-hosted platform on a repurposed 2018 laptop running immutable Fedora: eleven containers across seven Compose stacks serving a photo library, GPU-accelerated LLM inference, and browser game streaming to the living-room TV. Every service sits behind a custom Caddy build with a wildcard Let's Encrypt certificate, bound to loopback with nothing exposed to the internet, and reachable the same way off the LAN over Tailscale. Every problem is written up in an incident log.

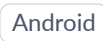
OTAMS

Tutoring Appointment App, SEG 2105 Team Project

 Sep 2025 – Dec 2025

 github.com/Diraclol/OTAMS

An Android app for booking tutoring sessions, built with a team of five. Students browse tutors and book time slots, tutors publish availability, and an administrator approves accounts. Firebase Auth and Firestore back it; it shipped in four graded iterations.

Spotify Mini Controller

Embedded Hardware Controller

 Sep 2025 – Dec 2025

A microcontroller-based controller for Spotify: MicroPython/C firmware reads a hardware button matrix and drives playback through the Spotify Web API in real time.